Andhra University College of Engineering

Mechanical Engineering | Course Code: ME301

Engineering Thermodynamics - Thermodynamics Laws, Cycles & Exergy Analysis

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Overview: First law steady-flow energy equation SFEE, Clausius inequality, Availability and Exe

I. UNIT-WISE SYLLABUS BREAKDOWN & HIGH-YIELD TOPICS

Unit 1: First Law of Thermodynamics

Work and heat transfers, First law for closed and steady-flow open systems SFEE, Enthalpy, Joule-Tho

Unit 2: Second Law & Entropy

Kelvin-Planck and Clausius statements, Carnot cycle, Clausius inequality, Entropy generation princip

Unit 3: Pure Substances & Vapor Cycles

P-v-T surfaces, Steam tables, Mollier diagram, Ideal Rankine cycle with Reheat and Regeneration

Unit 4: Gas Power Cycles

Air standard cycles: Otto, Diesel, Dual, and Brayton gas turbine cycles with thermal efficiency deri

II. HIGH-YIELD SEMESTER EXAMINATION QUESTIONS

1. Derive the thermal efficiency of an air-standard Diesel cycle in terms of compression ratio and cut-of
2. Formulate the steady-flow energy equation SFEE and apply it to a steam turbine and converging-divergin
3. Prove that the entropy of an isolated system always increases or remains constant during reversible pr

III. PRESCRIBED TEXTBOOKS & ACADEMIC REPOSITORIES

* Yunus A. Cengel, Michael A. Boles - Thermodynamics: An Engineering Approach McGraw-Hill
* P.K. Nag - Engineering Thermodynamics Tata McGraw-Hill 6th Ed

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